

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

20th March 2018

Subject ST8 – General Insurance: Pricing
Time allowed: Three Hours (14.45* – 18.00 Hours)

Total Marks: 100

INSTRUCTIONS TO THE CANDIDATES

1. *Please read the instructions inside the cover page of answer booklet and instructions to examinees sent along with hall ticket carefully and follow without exception.*
2. ** You have 15 minutes at the start of the examination in which you are required to read the questions. You are strongly encouraged to use this time for reading only, but notes may be made. You then have three hours to complete the paper.*
3. *You must not start writing your answers in the answer sheet until instructed to do so by the supervisor.*
4. *The answers are not expected to be any country or jurisdiction specific. However, if examples/illustrations are required for any answer, the country or jurisdiction from which they are drawn should be mentioned.*
5. *Attempt all questions, beginning your answer to each question on a separate sheet.*
6. *Mark allocations are shown in brackets.*
7. *Please check if you have received complete Question paper and no page is missing. If so, kindly get a new set of Question paper from the Invigilator.*

AT THE END OF THE EXAMINATION

Please return your answer booklet and this question paper to the supervisor separately. You are not allowed to carry the question paper in any form with you.

- Q. 1)** The claim numbers of an auto insurance portfolio follow Poisson (2) distribution and the claim amounts follow Pareto distribution with parameters $\alpha = 5$ and $\lambda = 1000$. There is an excess of 100 which applies to each claim.

Use simulation to generate two values from the aggregate loss amount distribution for the insurer, allowing for the excess. Use as many of the following values from a $U(0,1)$ distribution as you need.

0.27 0.82 0.44 0.66 0.81 0.51 0.02 0.66 0.98

[3]

- Q. 2)** List eight major restrictions that can be imposed by the regulators on the actions of a general insurance company.

[4]

- Q. 3)** You are the pricing actuary of an insurance company which needs to price an insurance policy for the business car insurance portfolio of a Car Rental company which has just come up for tender. The broker has provided aggregated experience data on claims, premiums and sums insured on the annual policies originating from the last five years up until 1/1/2018:

Policy Details				Motor Claims	
Policy Start Date	Total Premium (in millions)	Count of Cars	Total Sums Insured (millions)	Count	Amounts (in millions)
01/05/2013	61	7,647	3,476	1,984	48
01/05/2014	84	7,989	3,997	2,112	45
01/05/2015	68	8,393	3,464	2,439	105
01/05/2016	69	8,733	3,817	2,240	56
01/05/2017	56	8,466	3,909	924	24
Total	338	41,228	18,663	9,699	278

In addition, you have been informed that the new total sum insured for 2018 – when the policy gets due for renewal in May – will be 4,183 million and the total number of cars to be insured will be 8,546. Based on the claims information provided by the broker a member in your team has determined that the IBNR on the portfolio is 23 million and the current inflation rate is 6% per annum effective.

- i) State the advantages and disadvantages of using Frequency – Severity method of rating over Burning cost approach of rating. (2)
- ii) Stating the assumptions, Calculate the premium rate per Rs.1000 Sum Insured using Burning Cost approach. (3)
- iii) Discuss additional factors, including those relating to historical data that would need to be considered before finalizing the actual premium charged on this proposal. (3)

[8]

- Q. 4)** You are the pricing actuary of a medium-sized insurer writing mainly personal motor line of insurance via only two distribution channels, namely, individual agents and insurance brokers. Both the distribution channels sell motor insurance by quoting the rates from an online system of the insurer where rates are reviewed by you on half-yearly basis. Despite the continuous monitoring, the combined ratio of the portfolio is 120% as against the target of 105% or below.

Your analysis has revealed that the average cost per claim of individual agents is significantly higher than the insurance brokers.

- i)** Explain what you would investigate to find the reasons for this difference. (4)

After investigating, it has been inferred that the individual loss ratio of each channel is similar to the planned loss ratio yet the combined ratio of the company is much higher than planned ratio.

- ii)** Discuss what you would further investigate for this high combined ratio. (5)
[9]

- Q. 5)** Title is the legal way of saying that someone owns a right to something. For real estate purposes, title refers to ownership of the property, meaning that someone has the rights to use that property. It may be a partial interest in the property or it may be the full.

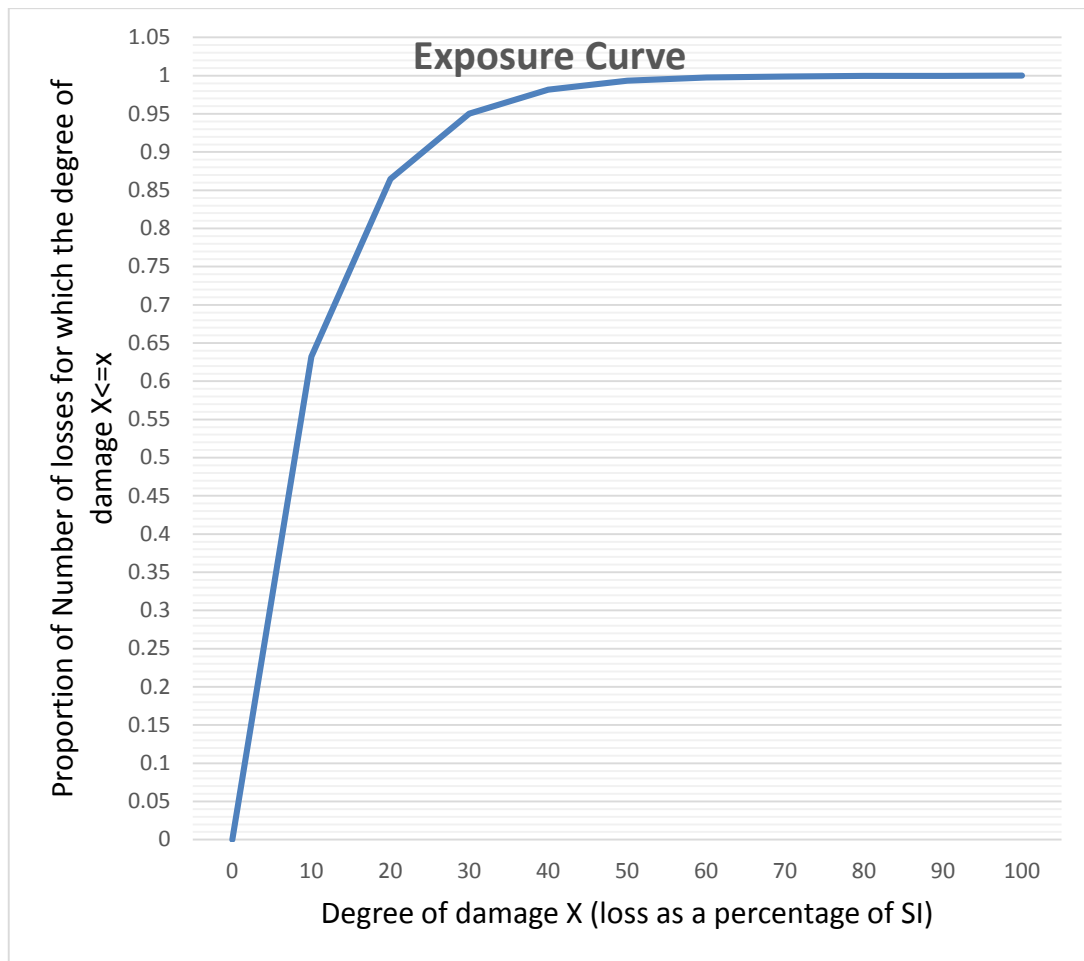
Title insurance is a form of indemnity insurance which insures against financial loss from defects in title to real property and from the invalidity or unenforceability of mortgage loans. In a mortgage loan, the property against which the loan is taken is usually taken as a collateral.

- i)** Outline the benefits, insured perils (or suitable examples of scenarios that can be covered), basis of sum insured, exposure measure, claim characteristics, risk and rating factors that you would consider for this Title insurance product. (8)

- ii)** State the reinsurance arrangements with reasons which would be most suitable for this product. (2)
[10]

- Q. 6)** **i)** The exposure curves, however they are expressed, are closely related to the original Loss distribution. Please state the important underlying assumption of exposure Curves which justifies the above statement. (1)
- ii)** State the reasons for not using original loss distribution instead of exposure curves. (1)

iii)



The above Figure 1 is an approximate representation of exposure curve for a property Fire class of business.

X-axis represents the degree of damage "X", where "X" represents loss as a percentage of SI.

Y-axis represents the number of losses for which the degree of damage $X \leq x$, as a percentage of the total number of losses.

- a) State how the exposure curve above is different from the standard exposure curves which show values on Y-axis that are proportional to the limited expected value (LEV) and the advantages of having such exposure curves. (3)

It is also given that the probability distribution function of x above follows an exponential distribution with $E(X) = 10\%$ and $\text{Var}(X) = 100\%$. You can also assume that the probability of a total loss is the value calculated as $P(X \geq 1)$ as per this distribution.

Table 1

Band Id	Lower Band	Upper Band	No. of risks	Premium	Total Sum Insured
			A	B	C
1	0	1,00,00,000	5000	5,00,00,000	30,00,00,00,000
2	1,00,00,000	5,00,00,000	1000	5,00,00,000	35,00,00,00,000
3	5,00,00,000	10,00,00,000	500	40,00,00,000	32,50,00,00,000
4	25,00,00,000	50,00,00,000	250	50,00,00,000	92,50,00,00,000
5	50,00,00,000	1,00,00,00,000	50	45,00,00,000	27,50,00,00,000

The table 1 provides the details of a property Fire class of business of a general insurance company. The expected loss ratio for the portfolio is 60%.

b) Using suitable assumptions, calculate the number of losses in each SI band. (2)

c) The insurance company has set a large loss threshold at Rs. 10 Cr. Calculate the number of large losses with loss size exceeding Rs. 10 Cr. (4)

[11]

Q. 7) i) State why the modelling data is divided into training data and validation data. (1)

ii) Briefly describe Gini Coefficient with respect to a pricing model. (2)

iii) Best insure is an insurance company operating in a developing country writing all lines of businesses. A lot of government and non-government agencies are spreading the awareness regarding the need of Home Insurance plan. The Company has also decided to offer home insurance policy with two sections:

- Structure insurance, which covers the building, and
- Home Contents insurance which covers the contents inside the insured home.

Discuss how you would use GLMs to arrive at a risk premium model for the Home contents sections of the policy. (10)

[13]

Q. 8) You are the chief actuary of a large insurance company specializing in niche product insurance. The company also underwrites personal accident insurance. One of the major selling product of the company is “Mobile Phone insurance” wherein the company covers the Accidental loss of or damage to the phone and the theft of the phone. The insurance is sold only when the policyholder buys the new mobile phone. The product is not available to the customer even after 1 day of purchase of the mobile phone.

There has been growing demand in the market for this product wherein customers are asking for this cover after the purchase of the product but within a certain specified time.

i) Discuss the various challenges in pricing this cover. (5)

ii) Outline how the risks stated in part (i) above can be mitigated? (3)

You are also reviewing your personal accident portfolio and the analysis reveals growing claims incidence rates in the recent periods. The incidence rates have increased mainly of young policyholders. Further investigations reveal that most of the accidents are due to mobile phone addiction e.g. using mobile phone while driving or walking on road, taking selfies at dangerous points like hill cliff etc.

To tackle this, you have advised the management to put ‘Accidents while using mobile phone’ as exclusion under the policy.

iii) Outline the points in favour and in against to this proposal. (5)

[13]

- Q. 9) i)** State the primary benefit of measuring premium rate changes for a portfolio. (1)
- ii) a)** Give some examples of definitions of premium rate used in the general Insurance Industry. (2)
- b)** State with reasons the definition among the examples which will be most applicable for risks like aircraft fleet. (2)
- iii) a)** Give some examples on the way the premium rate changes can be calculated. (2)
- b)** There was a small portfolio of only 100 liability policies written in 2015. Only 60 policies out of this portfolio renewed in 2016.

Out of these 60 renewed policies in 2016, there were three groups:

- A. In 2015, 20 policies had a SI limit of Rs. 20 lacs with a deductible of Rs. 1 lac with premium for each policy at Rs. 1500.
- B. In 2015, 20 policies had a SI limit of Rs. 20 lacs with a deductible of Rs. 2 lac with premium for each policy at Rs. 1300.
- C. In 2015, 20 policies had a SI limit of Rs. 20 lacs with a deductible of Rs. 5 lac with premium for each policy at Rs. 1000.

In 2016:

- The same set of policies under group A got renewed with deductible of Rs. 2 lacs instead of Rs 1 lac and Sum Insured of Rs. 20 lacs remaining the same, with a premium for each Policy at Rs. 1450 now.
- The same set of policies under group B got renewed with deductible of Rs. 5 lacs instead of Rs 2 lacs and Sum Insured of Rs. 20 lacs remaining the same, with a premium for each Policy at Rs. 1250 now.
- The same set of policies under group C got renewed with deductible of Rs. 5 lacs remaining the same but Sum Insured of Rs. 30 lacs instead of Rs. 20 lacs, with a premium for each Policy at Rs. 2000 now.

Assume all the premiums given above are excluding commission and management expenses. Also assume that appropriate loadings for commission and management expenses are applied to arrive at the final premium charged.

Please select a suitable method for calculating premium rate change and comment on the premium rate change of the renewed portfolio and its impact on the profitability in FY 16-17 compared to FY 15-16 assuming that the other margins (acquisition costs + management expenses) are fully covered.

The ILF table pertaining to the portfolio as at 2015 is given below:

Limit	ILF
1,00,000	1
2,00,000	1.432
3,00,000	1.717
4,00,000	1.927
5,00,000	2.092
10,00,000	2.6
15,00,000	2.877
20,00,000	3.058
30,00,000	3.288
40,00,000	3.431
50,00,000	3.53
1,00,00,000	3.771

(5)

- iv) Describe how the premium rate change calculations would change in question (iii)(b) above, if there has been a claims inflation of 5% annually from 2015 onwards and policy duration changes from annual to two years during the renewal. Assuming risks are uniform across the policy period. Rest of the details remain the same for the renewed portfolio except that the premiums are doubled during the renewal. State all the suitable assumptions.

(2)

[14]

- Q. 10) i) a)** Define reinstatements in the context of reinsurance contracts.

(1)

- b)** State the purpose and types of reinstatements.

(2)

- ii) Suppose there is an annual Excess of Loss treaty 200 Cr. xs 200 Cr, with the excess of loss treaty premium at Rs. 20 Cr. As per the treaty terms, there are two paid full reinstatements; 1st reinstatement at 50% of the XL premium and the 2nd reinstatement at 100% of the XL premium, pro-rata to the amount only.

Suppose there are five consecutive losses during the year covered by the Excess of Loss treaty. The loss details from the ground up against the five losses are given below:

1st Loss: Rs. 400 Cr.

2nd Loss: Rs. 300 Cr.

3rd Loss: Rs. 300 Cr.

4th Loss: Rs. 500 Cr.

5th Loss: Rs. 100 Cr

Please tabulate Recovery Amounts, Cover reinstated and Additional Reinstatement Premium against each loss from the Excess of Loss treaty assuming that there are no other reinsurance arrangements.

(4)

- iii) Suppose there is an annual CAT Excess of Loss treaty 200 Cr. xs 200 Cr with two free reinstatements and the CAT excess of loss treaty premium is Rs.40 Cr. Assume that every CAT loss is so large that it will always be more than Rs. 400 Cr. Also the frequency of the annual CAT event is assumed to follow Poisson distribution with mean of 0.1.

Calculate the discount which needs to be given on the current CAT excess of loss treaty premium to arrive at the new CAT excess of loss treaty premium if the two free reinstatements are now instead converted into paid reinstatements as follows:
1st reinstatement at 50% of the new Excess of Loss premium and the 2nd reinstatement at 100% of the new Excess of Loss premium, pro-rata to the amount only. You can ignore the timing of the payments for simplicity. (5)

- iv)** Please describe under what circumstances a reinsured should prefer to choose paid reinstatements instead of free reinstatements. (3)
[15]
